

CBSE Class 10 Mathematics — Triangles

Time Allowed: 3 Hours

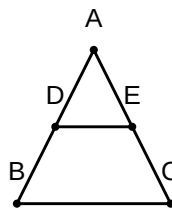
Maximum Marks: 80

General Instructions:

1. This question paper contains 24 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

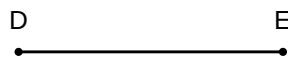
Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. In $\triangle ABC$, if $DE \parallel BC$ such that $AD = 2$ cm, $DB = 3$ cm, and $AE = 4$ cm, find the length of EC .



- (A) (A) 4 cm
(B) (B) 5 cm
(C) (C) 6 cm
(D) (D) 8 cm
2. If $\triangle ABC \sim \triangle PQR$ and $\frac{AB}{PQ} = \frac{1}{3}$, then the ratio $\frac{ar(\triangle ABC)}{ar(\triangle PQR)}$ is:
(A) (A) 1:9
(B) (B) 1:3
(C) (C) 3:1
(D) (D) 9:1
 3. Two poles of height 6 m and 11 m stand on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.
(A) (A) 12 m
(B) (B) 13 m
(C) (C) 14 m

- (D) (D) 15 m
4. In $\triangle ABC$, D and E are points on sides AB and AC respectively such that $AD = 3$ cm, $DB = 2$ cm, $AE = 4.5$ cm, and $EC = 3$ cm. Which of the following is true?
- (A) (A) $DE \parallel BC$
 (B) (B) $DE \parallel AB$
 (C) (C) $DE \parallel AC$
 (D) (D) DE is not parallel to any side
5. If $\triangle ABC \sim \triangle DEF$, $\angle A = 50^\circ$, and $\angle E = 70^\circ$, then $\angle C$ is:
- (A) (A) 50°
 (B) (B) 70°
 (C) (C) 60°
 (D) (D) 80°
6. Assertion (A): If $\triangle ABC \sim \triangle DEF$ and $AB = 2$ cm, $DE = 4$ cm, then the ratio of their areas is 1 : 4. Reason (R): The ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
7. Assertion (A): In $\triangle ABC$, if $DE \parallel BC$ with D on AB and E on AC , then $AD/DB = AE/EC$. Reason (R): Thales Theorem (Basic Proportionality Theorem) states that if a line is drawn parallel to one side of a triangle intersecting the other two sides, then it divides the two sides in the same ratio.

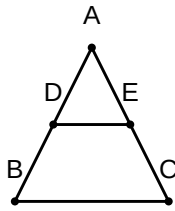


- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

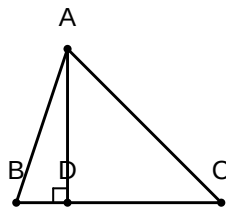
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
8. Assertion (A): Two triangles are similar if their corresponding angles are equal. Reason (R): If the corresponding sides of two triangles are in the same ratio, then the triangles are congruent.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): In $\triangle ABC$, if $\angle A = 90^\circ$ and $AD \perp BC$, then $\triangle ADB \sim \triangle CDA$. Reason (R): Two triangles are similar if two angles of one triangle are respectively equal to two angles of the other triangle.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
10. Assertion (A): The perimeters of two similar triangles are in the same ratio as their corresponding sides. Reason (R): If $\triangle ABC \sim \triangle PQR$, then $AB/PQ = BC/QR = CA/RP = (AB + BC + CA)/(PQ + QR + RP)$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

1. In $\triangle ABC$, $DE \parallel BC$ such that $AD = 3 \text{ cm}$, $DB = 4 \text{ cm}$, and $AE = 6 \text{ cm}$. Find the length of EC .

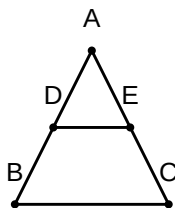


2. Two triangles $\triangle ABC$ and $\triangle DEF$ are similar. If $\text{Area}(\triangle ABC) = 64 \text{ cm}^2$ and $\text{Area}(\triangle DEF) = 121 \text{ cm}^2$, and $EF = 11 \text{ cm}$, find the length of BC .
3. In $\triangle PQR$, S and T are points on sides PQ and PR respectively such that $\angle PST = \angle PQR$. If $PS = 2 \text{ cm}$, $SQ = 3 \text{ cm}$, and $PT = 1.5 \text{ cm}$, find the length of TR .
4. A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.
5. In $\triangle ABC$, $AD \perp BC$. Prove that $AB^2 + CD^2 = AC^2 + BD^2$.



Section D: Long Answer (4-5 marks each)

1. In $\triangle ABC$, $DE \parallel BC$ such that $AD = (4x - 3) \text{ cm}$, $AE = (8x - 7) \text{ cm}$, $BD = (3x - 1) \text{ cm}$, and $CE = (5x - 3) \text{ cm}$. Find the value of x .



2. Two poles of height 6 m and 11 m stand on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.
3. In $\triangle ABC$, $\angle A = 90^\circ$ and $AD \perp BC$. If $AD = 8$ cm and $BD = 4$ cm, find the length of CD .
4. The perimeters of two similar triangles are 30 cm and 20 cm respectively. If one side of the first triangle is 9 cm, find the length of the corresponding side of the second triangle.
5. In $\triangle ABC$, P and Q are points on sides AB and AC such that $PQ \parallel BC$. If $AP = 3$ cm, $PB = 2$ cm and $AQ = 2.7$ cm, find AC .

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6. Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. Use this theorem to find the length of EC in *riangle* ABC where $DE \parallel BC$, $AD = 3$ cm, $DB = 4$ cm, and $AE = 6$ cm.

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7. In $\triangle ABC$, $\angle A = 90^\circ$ and $AD \perp BC$. Prove that $AD^2 = BD \times CD$. If $BD = 4$ cm and $CD = 9$ cm, calculate the length of AD .
8. Two poles of heights a meters and b meters are p meters apart. Prove that the height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is given by $\frac{ab}{a+b}$ meters.

9. Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides. Using this, find the ratio of the areas of two similar triangles if their sides are in the ratio 4 : 9.